Lesson 13: A Hit That Reacts

GDevelop

VIDEO GAME CREATION

This week: getting hit finally LOOKS like getting hit. A brief flash, then a small explosion, before the enemy disappears.

Open your own [YourName]Game project from Lesson 12 - don't start a new one.

Important: your score, lives, wave, and win/lose logic should all still work exactly as before. Today only changes how the enemy's removal LOOKS, not when your game-state actually changes.

Setup

1. Open your [YourName]Game project from Lesson 12.
2. Add a new Sprite object named Explosion, using explosion_placeholder.png as its image.
3. In the Extensions panel, install the official "Flash" extension if it isn't already installed (one click) - this makes the Flash behavior available to add to objects, but doesn't attach it to anything yet.
4. Select Enemy, open its Behaviors tab, click Add a behavior, search "Flash," and Apply. Flash visibility (blink) is an action that belongs to this behavior, not a general action - it won't show up in Enemy's action list until this step is done, even though the extension itself is already installed.

Level 1 - Minimal (MVP)

On every hit, the enemy briefly flashes, and a small independent explosion effect appears and cleans itself up. Score, lives, wave, and win/lose logic are all unaffected.



Everything in the top box happens the instant the hit is detected. Only the enemy's own removal - the very last step - waits for the pause to finish.

5. On the bullet-enemy collision event, right after the existing score/sound actions, add: Create object Explosion at Enemy.X(), Enemy.Y() (spawns immediately, independent of everything else) and Flash visibility (blink) of Enemy, Duration 0.3 seconds - the Flash behavior's blink action (now available since the behavior was added to Enemy in Setup), which toggles the Enemy's visibility on/off for that duration. This is deliberately the blink action, not the Flash behavior's separate color-tint action: a color tint multiplies against the Enemy's existing color, and multiplying any color by white leaves it completely unchanged - it wouldn't actually flash on a colored sprite like ours. Blink works regardless of sprite color, since it's just toggling visibility.
6. Add a Wait 0.1 seconds action right after the flash action, still inside the same event.
7. Move Delete Enemy to come AFTER the Wait action - everything else in the event (Score, EnemiesRemaining, Play a sound, Delete Bullet) stays exactly where it was, unchanged.
8. Add a separate event: Condition: Explosion has existed for more than 0.3 seconds -> Action: Delete Explosion, so explosions clean themselves up automatically.
9. Click Preview and play through a full round - each hit should flash the enemy briefly, and a small burst should appear at that spot for a moment.

Highlighted lines are new this week; everything else is code you already have.

```
Bullet-Enemy collision event (revised):
Condition: Bullet is in collision with Enemy
Action: Score = Score + 10
Action: EnemiesRemaining = EnemiesRemaining - 1
Action: Play a sound (sfx_hit.wav)
```

```
Action: Create object Explosion at Enemy.X(), Enemy.Y()
Action: Flash visibility (blink) of Enemy, Duration 0.3
Action: Delete Bullet
Action: Wait 0.1 seconds
Action: Delete Enemy

Separate event (explosion self-cleanup):
Condition: Explosion has existed for more than 0.3 seconds
Action: Delete Explosion
```

Level 2 - Complete (Alpha/Beta)

The flash and explosion play correctly on every single hit, with no leftover explosions, no enemies stuck mid-flash, and no change to score/lives/wave timing.

- Play a full round: every single hit shows a flash and an explosion, with no enemy left stuck mid-flash.
- Confirm score, wave-clear, and win/lose all still happen at exactly the moments they did last week - EnemiesRemaining changes immediately, only Delete Enemy is delayed.
- Confirm Explosion objects never pile up - each one deletes itself via its own timed event, with no help from the collision event.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: add the official Tween behavior to Explosion, and tween its scale from 0.5 to 1.5 over 0.3 seconds for an expanding-burst look instead of a static image.
- Visual: try the Flash extension's separate color-tint-flash action instead of the blink action, and see what happens on your Enemy's actual sprite color - a multiply-based tint can only darken/mix with a sprite's existing color, so a "white" tint on a colored sprite comes out looking like the original color, not white. It's still worth trying (and a reasonable stretch goal), but pick a color noticeably different from your Enemy's own for a result you can actually see, rather than white specifically.
- Structure: notice Explosion knows nothing about score, lives, or waves - it's a fully self-contained object, the same discipline as this course's bullets and enemies.

If Something's Wrong

- Enemy vanishes instantly, no flash visible: confirm Wait 0.1 seconds was actually added, and Delete Enemy comes AFTER it, not before.
- Explosion never appears: confirm the Flash extension is actually installed, and that Create object Explosion references the correct object name.
- Explosions pile up and never disappear: confirm the separate self-cleanup event's "has existed for more than" condition is actually targeting the Explosion object, not a typo'd name.
- Enemy doesn't flash at all, even though the extension is installed: installing the Flash extension only makes the behavior available to add - it doesn't attach it to any object automatically. Confirm the Flash behavior was actually added to Enemy specifically (Enemy > Behaviors tab).

Want More?

Student Resource List > GDevelop section, or wiki.gdevelop.io